

Late Gate — Parametric Flight-Delay Stubs: Data Landscape, Pricing Model, and Risk Controls

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Pricing lock: Takeoff $\pi = \$14$; Arrival $\pi = \$9$; $B(30/45/60) = \$100 / \$150 / \$200$ (both products). Fixed premium; scaled payout. τ internal only.

ABSTRACT / OVERVIEW

Late Gate offers travelers limited-inventory **parametric stubs** on two outcomes per flight: **Takeoff delay** and **Arrival delay**. The traveler selects a delay-buffer tier; the stub pays a fixed cash amount if observed schedule delay meets or exceeds an internal threshold. The house is the sole counterparty in the MVP.

This paper (1) surveys public and commercial flight-performance data sources; (2) maps gate vs runway delay definitions to Late Gate products; (3) reports empirical US delay rates at 15/30/45/60 minutes from Bureau of Transportation Statistics (BTS) microdata; (4) specifies a pricing model with load λ and house EV; (5) shows that the **locked Eng premiums are underwater at US pool-average probabilities** and therefore must be paired with **selective underwriting**; (6) defines inventory, adverse-selection, correlation, and settlement controls; and (7) notes traveler-safe wording and a future depositor-pool sketch.

Key empirical result (BTS Reporting Carrier OTP, Jan + Jul 2024, operated non-cancelled non-diverted, $n \approx 1.14M$):

Product proxy	P(≥ 30 min)	P(≥ 45)	P(≥ 60)
Departure / Takeoff (DepDelay)	18.33%	13.61%	10.50%
Arrival (ArrDelay)	18.66%	13.83%	10.69%

At locked π and B, expected payout on a **random** US flight exceeds premium (house EV < 0). Positive house EV is restored on a **clean demo / selectively underwritten** book (e.g. $\hat{p}_{30/45/60} \approx 8\%/5\%/3.5\%$).

1. Data landscape

1.1 BTS Airline On-Time Performance (primary free US prior)

- **Coverage:** US domestic nonstop flights of reporting carriers ($\geq 0.5\%$ of domestic scheduled passenger revenues).
- **Fields:** Scheduled/actual gate times, DepDelay / ArrDelay, 15-minute flags, delay-cause minutes, taxi, wheels-off/on, cancellations, diversions, O&D, distance, carrier.
- **Cadence:** Monthly; typically lagged ~ 1 – 2 months.
- **Access / license:** Free public download (TranStats prezipped ZIPs). US government statistical data.
- **URLs:**
 - https://transtats.bts.gov/DatabaseInfo.asp?DB_URL=&QO_VQ=EFD
 - https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ&QO_fu146_anzr=b0-gvzr
- Prezip pattern:
<https://transtats.bts.gov/PREZIP/On Time Reporting Carrier On Time Performance 1987-present YYYY M.zip>

1.2 EUROCONTROL CODA (Europe)

- All-causes delay and punctuality for ECAC IFR traffic; monthly briefs + annual Digest.
- 2024: $\sim 72.4\%$ arrivals within 15 min of STA ($\approx 27.6\% > 15$); avg delay ≈ 17.5 min/flight.
<https://www.eurocontrol.int/publication/all-causes-delays-air-transport-europe-annual-2024>
- 2025 review: arrival punctuality 76.1%; departure 70.1%; avg departure delay 14.6 min.
<https://www.eurocontrol.int/sites/default/files/2026-01/eurocontrol-european-aviation-overview-20260123-2025-review.pdf>

- Monthly band charts publish Del 16–30 / 31–60 / >60 shares (e.g. Sep 2025 >30 min: dep 18.3%, arr 16.5%).
<https://www.eurocontrol.int/sites/default/files/2025-10/eurocontrol-all-causes-delay-to-air-transport-september-2025.pdf>

1.3 Cirium Sky API (commercial — settlement candidate)

Global status, gate/runway delays, historical status (contract). Trial / PAYG / contract. Terms restrict passenger-rights claim use.
<https://developer.cirium.com/apis/cirium-sky-api/subscriptions>

1.4 OAG Flight Status (commercial)

Near-real-time gate & runway times, delays, cancellations, diversions; API / alerts / Snowflake. Quote-based.
<https://www.oag.com/flight-status-data>

1.5 FAA ASPM / ASQP

US airport and carrier metrics; gate out/in, wheels off/on. Aggregates partly public; flight-level often login-gated.
[https://aspmhelp.faa.gov/index/Aviation_System_Performance_Metrics_\(ASPM\).html](https://aspmhelp.faa.gov/index/Aviation_System_Performance_Metrics_(ASPM).html)
https://aspmhelp.faa.gov/index/ASQP_Definitions_of_Variables.html

1.6 OpenSky Network

ADS-B tracking — **not** CRS schedule delay. Research free under terms; commercial / operational use requires license.
<https://opensky-network.org/about/terms-of-use>

1.7 Blockers

Cirium/OAG paywalled (needed for live settlement). BTS Excel summary tables often blocked to datacenter IPs (prezip CSVs work). OpenSky unsuitable as sole settlement oracle for schedule delay.

2. Delay definitions and product mapping

Industry metric	Definition	Late Gate product
Gate departure delay	Actual gate-out – scheduled dep (BTS DepDelay)	Takeoff delay
Wheels-off delay	Runway liftoff – schedule (or vs gate-out + taxi)	Not primary trigger
Gate arrival delay	Actual gate-in – scheduled arr (BTS ArrDelay)	Arrival delay
Wheels-on / block	Touchdown / in-block variants	Not primary
BTS “late flight”	≥15 minutes gate late	Reporting standard only

BTS: gate departure is parking-brake release / door close (with specified gate-return rules).
https://transtats.bts.gov/DatabaseInfo.asp?DB_URL=&QQ_VQ=EFD

Recommendation: Settle Takeoff on **gate-out vs published STD**; Arrival on **gate-in vs published STA**. Traveler-facing “Takeoff delay” means departure lateness from schedule, not wheels-up. **Never display τ minutes in UI.**

3. Empirical rates

3.1 US ≥15 minutes (published)

BTS Table 1A / TranStats (reporting carriers): late arrivals/departures ≈ **20.3%** (2023), ≈ **20.4–20.5%** (2024). Marketing-carrier arrival delay ≈20.2% (2025 chart series).
Sources: TranStats home drill / Marketing Annual; Table 1A xlsx on bts.gov (search-indexed extracts).

3.2 US ≥30/45/60 — this study (BTS microdata)

Method: Download TranStats Reporting Carrier OTP for **2024-01** and **2024-07**; filter Cancelled=0, Diverted=0; compute share with DepDelay / ArrDelay ≥ τ.

Period	Dep ≥30	Dep ≥45	Dep ≥60	Arr ≥30	Arr ≥45	Arr ≥60
Jan 2024	15.47%	11.27%	8.57%	15.94%	11.54%	8.78%
Jul 2024	20.77%	15.61%	12.15%	20.99%	15.80%	12.32%
Weighted	18.33%	13.61%	10.50%	18.66%	13.83%	10.69%

Artifacts: data/empirical_delay_rates_*.csv, figures/empirical_delay_cdf.png.

Caveats: Two months only; summer worse; cancellations/diversions excluded from CDF (need separate rules); not conditioned on purchase timing (adverse selection).

3.3 Heterogeneity (Jul 2024 departures, p≥30)

Split	Low	High
Airline	HA 6.6%	AA 28.9%
Dep time block	0600–0659 7.6%	2000–2059 35.6%
Origin (top volume)	SEA 14.5%	CLT 35.9%
DOW	Thu 17.5%	Fri 25.7%

Full tables: data/empirical_delay_splits_jul2024.csv.

3.4 Europe (published aggregates)

EUROCONTROL: >15 min arrivals ~24–28% depending on year; monthly >30 min often mid-teens (e.g. Dec 2024 arr 13.7%; Sep 2025 arr 16.5%). Exact annual EU p≥45/60 not published as a single scalar in materials reviewed — treat as **unknown** pending CODA microdata or Cirium/OAG.

4. Pricing model

4.1 Contract

For product $k \in \{TAKEOFF,ARRIVAL\}$ and traveler-selected internal threshold $\tau \in \{30,45,60\}$:

$$Payout = B(\tau) \cdot 1_{\{D_k \geq \tau\}}, \quad B(30)=100, \ B(45)=150, \ B(60)=200$$

Premium π_k is **fixed** per product (not per τ):

$$\pi_{TAKEOFF} = 14, \quad \pi_{ARRIVAL} = 9$$

Expected payout: $E[payout] = p_k(\tau) \cdot B(\tau)$ where $p_k(\tau)=P(D_k \geq \tau)$.

4.2 Load and house EV

$$\pi = \lambda \cdot E[payout] = \lambda \cdot p \cdot B \quad \Rightarrow \quad EV_{house} = \pi - p \cdot B$$

Target $\lambda = 1.45$ on the underwritten book (includes ~20pp adverse-selection / ops buffer vs pure fair odds $\lambda=1$). Traveler “fair odds” relative to estimated p means transparent load: fair premium would be $p \cdot B$; traveler pays $\lambda \cdot p \cdot B$ when π is set by underwriting, or equivalently only flights with $p \leq \pi/(\lambda \cdot B)$ are sold at locked π .

4.3 Validation of Eng lock vs data

Pool-average p (weighted Jan+Jul 2024):

Product	τ	p	$E[\text{payout}]$	π	House EV	Implied λ
Takeoff	30	18.33%	\$18.33	\$14	−\$4.33	0.76
Takeoff	45	13.61%	\$20.41	\$14	−\$6.41	0.69
Takeoff	60	10.50%	\$21.00	\$14	−\$7.00	0.67
Arrival	30	18.66%	\$18.66	\$9	−\$9.66	0.48
Arrival	45	13.83%	\$20.75	\$9	−\$11.75	0.43
Arrival	60	10.69%	\$21.38	\$9	−\$12.38	0.42

Decision: Retain Eng π/B as **locked baseline** for product UX and Rails constants. **Do not** sell at these prices on pool-average risk. Enforce λ via flight-level $\hat{p}$ gates.

4.4 Underwriting gates ($p_{\max}$)

At $\lambda=1.45$: $p_{\max} = \pi / (\lambda B)$.

Product	τ	$p_{\max}$	Break-even p ($\lambda=1$)
Takeoff	30	9.66%	14.00%
Takeoff	45	6.44%	9.33%
Takeoff	60	4.83%	7.00%
Arrival	30	6.21%	9.00%
Arrival	45	4.14%	6.00%
Arrival	60	3.10%	4.50%

4.5 Clean demo prior (recommended)

Assume selectively listed flight with $\hat{p}=\{8\%,5\%,3.5\%\}$ at $\{30,45,60\}$:

Product	τ	$E[\text{payout}]$	House EV	λ
Takeoff	30	\$8.00	+\$6.00	1.75
Takeoff	45	\$7.50	+\$6.50	1.87
Takeoff	60	\$7.00	+\$7.00	2.00
Arrival	30	\$8.00	+\$1.00	1.12
Arrival	45	\$7.50	+\$1.50	1.20
Arrival	60	\$7.00	+\$2.00	1.29

Arrival at \$9 is tight ($\lambda \approx 1.1\text{--}1.3$ on this prior). Prefer morning / historically clean flights for Arrival inventory.

4.6 Alternatives if Eng reopens pricing (not locked)

For **unselective** US pool book at $\lambda=1.45$ and current B: $\pi \approx$ **\$27–\$31** both products. Or keep π and reduce B to $\sim$ \$30–\$55 (poor UX). **Recommendation:** keep locked π/B + selection for demo; revisit risk-based π later.

5. Risk controls

5.1 Inventory caps (FOMO)

- Default **10** stubs per flight per product (demo 5; max 20).
- Max payout exposure $\approx$ inventory $\times$ \$200 (default **\$2,000** / flight $\times$ product).

5.2 Adverse selection — operational codes

Code	Rule
HOT	Freeze new sales if delay already developing, weather watch, or soft $\hat{p}$ warn
CUTOFF	No sales inside 4h of STD (demo 6h)
FULL	Inventory exhausted
DUPLICATE	Max 1 open stub per user per flight×product
UNDERWRITE_REJECT	$\hat{p} > p_{\max}$ for product× τ
EXPOSURE_CAP	Portfolio / cluster limit hit
CANCELLED	MVP: refund premium
DIVERTED	Arrival: pay max B or review

5.3 Correlation / weather / portfolio

- Caps: airport-day \$25k; airline-day \$40k; book-day \$150k; cluster \$15k (max simultaneous payout exposure).
- Weather-watch airports: remaining capacity $\times$ **0.35**.
- Prefer diversifying away from single-hub evening banks.

5.4 Why controls matter for EV

Purchase timing correlates with information about delay. CUTOFF + HOT shrink adverse selection so realized p on sold stubs approaches (or stays under) underwriting $\hat{p}$. Without them, locked π is insufficient even on “clean” scheduled flights.

6. Settlement

1. **Oracle:** Production — Cirium or OAG (or airline direct). MVP — partner status API; BTS for offline backtest only (lag).
2. **Metric:** Takeoff = gate-out delay; Arrival = gate-in delay vs published times at purchase lock.
3. **Observe:** After actual event or scheduled time + τ + **30 min** buffer.
4. **Auto-pay** if high-confidence source and $D \geq \tau$; else hold.
5. **Dispute window:** 72 hours.
6. **Cancelled:** refund π (MVP). **Diverted:** Arrival special case.

7. Sensitivity

7.1 House EV vs p at locked π

See figures/house_ev_sensitivity.png. Zero-crossing at break-even $p=\pi/B$. Pool p_{30} (≈ 18 – 19%) lies deep in the loss region for both products.

7.2 Inventory $\times \lambda$

Inventory	Takeoff premium income	Max exposure ($\tau=60$)	Arrival premium income
5	\$70	\$1,000	\$45
10	\$140	\$2,000	\$90
20	\$280	\$4,000	\$180

House EV scales with $\sum_i (\pi - p_i B_i)$ over sold stubs — negative if p_i uncontrolled.

7.3 Traveler fair-odds comparison (clean demo prior)

For Takeoff $\tau=30$: fair premium $\approx$ \$8.00; traveler pays \$14 $\rightarrow$ load 1.75 (pays \$6 above actuarial fair for that prior). For Arrival $\tau=30$: fair $\approx$ \$8.00; pays \$9 $\rightarrow$ load 1.12 (thin). Communicate value as **certainty of payout rules + limited inventory**, not “cheap odds.”

8. Future depositor pool (optional)

MVP: house sole counterparty. Future: depositors post capital; after traveler payouts and ops, residual edge $(\lambda-1) \cdot E[payout]$ can be shared, e.g. 40% house ops / 50% depositor yield / 10% reserve. Depositors **inherit house edge only after** underwriting and reserves — they do not remove the need for $p_{\max}$ gates. No user LP in current product.

9. Traveler-safe wording notes

Avoid (UI)	Prefer
insurance, policy, claim, premium	stub, locked price, payout
gamble, bet, odds	delay cushion, buffer tier
show “30/45/60 minutes” as τ	Short / Medium / Long buffer
house edge, λ , $\hat{p}$	Limited stubs on this flight

Paper/actuarial language is fine internally and in this document; traveler surfaces should stay assistance/ancillary-protection tone (consistent with Hopper-style “disruption assistance” framing in market), without claiming regulated insurance unless licensed.

10. References

1. BTS Airline On-Time Performance Database — https://transtats.bts.gov/DatabaseInfo.asp?DB_URL=&QO_VQ=EFD

2. BTS Download (Reporting Carrier) — https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ&QO_fu146_anzr=b0-gvzr

3. BTS TranStats annual charts — <https://www.transtats.bts.gov/homedrillchart.asp>

4. BTS Marketing Carrier Annual — https://www.transtats.bts.gov/Marketing_Annual.aspx

5. BTS on-time gallery — <https://www.bts.gov/browse-statistical-products-and-data/info-gallery/percent-time-flight-departures-and-arrivals>

6. EUROCONTROL CODA Digest 2024 — <https://www.eurocontrol.int/publication/all-causes-delays-air-transport-europe-annual-2024>

7. EUROCONTROL Sep 2025 delay brief PDF — <https://www.eurocontrol.int/sites/default/files/2025-10/eurocontrol-all-causes-delay-to-air-transport-september-2025.pdf>

8. EUROCONTROL 2025 review PDF — <https://www.eurocontrol.int/sites/default/files/2026-01/eurocontrol-european-aviation-overview-20260123-2025-review.pdf>

9. FAA ASQP definitions — https://aspmhelp.faa.gov/index/ASQP_Definitions_of_Variables.html

10. FAA ASPM overview — [https://aspmhelp.faa.gov/index/Aviation_System_Performance_Metrics_\(ASPM\).html](https://aspmhelp.faa.gov/index/Aviation_System_Performance_Metrics_(ASPM).html)

11. Cirium Sky subscriptions — <https://developer.cirium.com/apis/cirium-sky-api/subscriptions>

12. OAG Flight Status — <https://www.oag.com/flight-status-data>

13. OpenSky terms — <https://opensky-network.org/about/terms-of-use>

14. Wang et al. delay CCDF (arXiv) — <https://ar5iv.labs.arxiv.org/html/1701.05556>

15. AXA HK Express parametric — <https://www.axapartners.com/en/page/axa-hk-express-flight-delay-parametric-protection>

16. Cover Genius Delay Valet — <https://covergenius.com/solutions/delay-valet/>

17. Blink Flight Disruption — <https://blinkparametric.com/blink-parametric-platform/blink-flight-disruption/>

18. Hopper / OAG case study — <https://www.oag.com/hopper-case-study>

19. Late Gate empirical outputs — `/workspace/late-gate-docs/data/`, `/workspace/late-gate-docs/figures/`

Appendix A — Rails constants

See `rails-pricing-constants.md` and `rails-pricing-constants.json` for drop-in thresholds, premiums, payouts, inventory, λ , refusal codes, and exposure limits.

Appendix B — Assumptions (labeled)

- **A1:** Settlement equals BTS-style gate delay (assumption until Cirium/OAG wired).
- **A2:** Sold-stub p equals historical flight-level $\hat{p}$ if CUTOFF/HOT enforced (optimistic without live telemetry).
- **A3:** Single-tier payout (not stacked 30+45+60).
- **A4:** Cancellations refund π (not a delay win).
- **A5:** Clean demo prior 8/5/3.5% is a **scenario**, not a national statistic.
- **A6:** Full-year 2024/2025 microdata not fully ingested — expand before production rate cards.

Late Gate — Exec Summary (Mikail)

Date: 12 Sep 2026 (ET)
Ask: Data-backed parametric stub pricing for Takeoff & Arrival delay products.

Locked Eng baseline (demo / Rails)

	Takeoff delay	Arrival delay
Premium π	\$14	\$9
Payouts B	\$100 / \$150 / \$200 at internal 30 / 45 / 60 min	same
Structure	Fixed premium; traveler picks one buffer tier; payout scales	same
Inventory	Limited stubs / flight / product (default 10; demo 5)	
Counterparty	House only (no user LP yet)	

τ stays **internal** — UI never shows “30/45/60 minutes” as underwriting jargon.

What the data says

From **BTS** flight-level On-Time Performance (Reporting Carrier), **Jan + Jul 2024** (~1.14M operated flights):

	p(delay ≥ 30)	≥ 45	≥ 60
Departure (Takeoff proxy)	18.3%	13.6%	10.5%
Arrival	18.7%	13.8%	10.7%

US industry ≥ 15 min late remains ~**20–21%** (BTS 2023–2025). Europe is similar or worse on >15 (EUROCONTROL 2024 arrival punctuality ~72% $\rightarrow$ ~28% >15).

Delays concentrate by **airline, airport, time-of-day, day-of-week** (e.g. Jul 2024 morning dep $p_{\geq 30}$ ~8%; evening ~35%; HA ~7% vs AA ~29%).

Critical pricing finding

At pool-average probabilities, Eng’s locked π /B loses money (house EV deeply negative; implied $\lambda \approx 0.4\text{--}0.8$).

Product	Example $\tau=30$	E[payout] at pool p	π	House EV
Takeoff	$18.3\% \times \$100$	\$18.33	\$14	–\$4.33
Arrival	$18.7\% \times \$100$	\$18.66	\$9	–\$9.66

We keep Eng numbers as the locked baseline for UI/demo/Rails constants, and make the book work via **selective underwriting**:

- Only list stubs when flight-level $\hat{p}$ is below gates (Arrival is stricter than Takeoff).
- Target load $\lambda \approx 1.45$ on the *underwritten* book.
- Clean demo prior $\hat{p} \approx 8\% / 5\% / 3.5\% \rightarrow$ Takeoff EV $\approx$ **+\$6 to +\$7**; Arrival EV $\approx$ **+\$1 to +\$2** per stub.

If Eng later reopens prices for a *random-flight* book, fair π at $\lambda=1.45$ & current B is roughly **\$27–\$31** per product — or cut B sharply. Prefer **selection + locked π** for the demo.

Top 5 risk controls

1. **UNDERWRITE_REJECT** / **p_max gates** — don't sell high- $\hat{p}$ flights.
 2. **CUTOFF** — stop sales before STD (4–6h).
 3. **HOT** — freeze sales when delay/weather develops.
 4. **Inventory + portfolio caps** — FOMO stubs; airport/airline/day exposure limits; weather multiplier 0.35.
 5. **DUPLICATE** — one stub / user / flight / product.
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Settlement (MVP)

- **Takeoff** = gate-out vs schedule; **Arrival** = gate-in vs schedule (matches BTS DepDelay / ArrDelay).
 - Observe with buffer; 72h dispute window.
 - Live settlement needs **Cirium/OAG** (paywalled); BTS is free but lagged (great for priors/backtest).
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Traveler copy

Avoid “insurance / gambling / odds.” Prefer stub, payout, delay cushion, limited inventory. Paper may use actuarial language; UI must not.

Deliverables

File	Path
Technical paper	/workspace/late-gate-docs/technical-paper-flight-delay-pricing.md
Research notes	/workspace/late-gate-docs/research-notes.md
Rails tables	/workspace/late-gate-docs/rails-pricing-constants.md
Rails JSON	/workspace/late-gate-docs/rails-pricing-constants.json
Data & figures	/workspace/late-gate-docs/data/, figures/

Bottom line: Ship Eng's **\$14 / \$9** and **\$100/\$150/\$200**, but **only on clean flights**. That is how house EV stays > 0 without abandoning the locked traveler-facing numbers.

Rendered from technical-paper-flight-delay-pricing.md and exec-summary.md. Prefer print/academic HTML or PDF; not a product website.